

Investigating the Synonyms of Conversational Agents to Aid Cross-Disciplinary CA Research

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ABSTRACT

With the advent of artificial intelligence, the CHI community has regained a large interest in Conversational Agents (CAs). Designing CAs involves cross-disciplinary efforts, such as that of computer science (e.g., language models), psychology (e.g., cognition and emotions), linguistics (e.g., conversation design), or communication (e.g., trust, communication theory). However, CAs are named differently depending on the discipline and purpose (e.g., chatbot, embodied avatar, virtual butler). This divergence of vocabulary on CA brings challenges to researchers and designers in researching the full landscape of CA literature and sharing cross-disciplinary knowledge. We performed bibliometric and qualitative analyses to systematically assess divergent terms used for CA nomenclatures. We present 54 CA-terms and how these terms are used differently depending on the discipline and design characteristics (e.g., voice vs. non-verbal vs. text-based). Our work contributes to helping CA researchers effectively review the CA literature and build on each other's work for novel cross-disciplinary CA research.

CCS CONCEPTS

• General and reference → Reference works; General literature; • Computing methodologies → Intelligent agents.

KEYWORDS

conversational agent, chatbot, bibliometric analysis, co-citation analysis

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1 INTRODUCTION

Conversational agents (CAs) [61], interacting with users in natural languages, like Apple Siri, Microsoft Cortana, and Amazon Alexa, have been applied in different contexts including health and fitness, business and finance, and education [91]. Developing usable and

intelligent CAs involve multiple disciplinary knowledge, including psychology, computer science, communication, and human-computer interaction. Because of this cross-disciplinary nature of CAs, active collaboration and sharing of research results across disciplines will be key to advancing CA research. Researchers call CA in many different ways including chatbot [3], dialogue system [12], conversational interface [12], intelligent agent [46], conversational assistant [9], virtual companion [69], avatar [69], etc. The website chatbots.org listed 161 terms that they believed were synonyms adopted by the industry to refer to “humanlike conversational AI”, including virtual human, conversational agent, chatbot and so on.

Literature in CA research has a long history involving researchers from diverse disciplines, which can be traced back to the 1960s when chatbot ELIZA was developed to mimic a psychotherapist [99]. With the advances in artificial intelligence (AI), natural language processing (NLP), and cognitive systems, CAs have progressed immensely in the design and development [16]. CAs' functions and applications vary greatly from robot-based CAs for education [29] to embodied CAs for therapy [83] to disembodied CAs for customer service [1].

A challenge here is that CA research is scattered across different areas and disciplines that generally do not overlap in publication venues with distinct agendas for CA research. For instance, in investigating ‘virtual avatars’, the field of communication identified various factors in how people perceive artificial agents’ emotions and their tasks [59], while the engineering field explored haptic interaction with virtual avatars [31]. Audience of these publication venues may not overlap and researchers need to dive across different publication venues to understand the achievements of virtual avatars and even may inform their findings through another different venue after their implementation and evaluation. Such cross-disciplinary barriers result in the inconsistent use of CA-terms and their usage in different research areas. This could further hinder the knowledge sharing of CA research among multiple disciplines.

As the CHI community continues to engage in CA research, we should have a shared vocabulary for describing the various characteristics of CA and build cross-disciplinary perspectives and knowledge on CA. Accordingly, our research questions include:

- RQ1: What are the different terms used to refer to the same underlying concept of conversational agents in the literature?
- RQ2: How do terms on CA and their usage differ depending on their design characteristics and research areas?

To address these questions, using 5,047 papers on CA, we performed bibliometric analysis to identify possible CA-terms

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(Terms used to denote CAs) and examined their usage context by qualitatively analyzing the content of the literature. The findings inform CA researchers with shared vocabulary to search the literature and build on each other's work. Our work will serve as a starting point for researchers to avoid duplicate work, connect fragmented research, and facilitate cross-disciplinary knowledge sharing.

2 RELATED WORK

McTear showed there is little consistency in the use of these various CA-terms when describing CAs in the research literature and in the media [71]. Some researchers attempted to distinguish various CA-terms by their functions in carrying out tasks [69] or by task-oriented or entertainment-oriented conversations [51], while others used these terms interchangeably [4].

With such inconsistency in the language around CA, challenges emerge around effective literature search. Previous systematic reviews on CA show various ways they applied search queries to find CA literature. For example, a study on the taxonomy of social cures for CAs applied 9 CA-terms e.g., “(conversational OR virtual OR digital) AND (agent* OR assistant*)” OR chatbot* OR chatterbot* OR chatterbox for finding CA research [32]. A review on dyadic CAs [103] used 15 CA-terms. Rheu et al.'s systematic review paper used 29 CA-terms in their search queries [86], and a study on CAs for mental health used two terms—conversational agent or chatbot [95].

Existing reviews on CAs also defined various criteria for screening and filtering CA literature based on their specific goals of the reviews [54, 61, 67, 73, 86, 93, 103]. Per existing CA reviews, some researchers focused on the domains such as health [54, 61, 73, 93, 95] and commerce [67], and some researchers interested in special user groups like children [35], while others investigated specific characteristics of CAs like text-based CAs [85], voice-based CAs [35], social characteristics [17], social cues [32], personalization [54], trust building [86], dyadic and polyadic interactions through CAs [103], and affordances [75].

To improve the quality and efficiency of literature collection, screening, and filtering, the search strategies can be further expanded by explaining how these terms were selected over others, given how search strategies lead to different collections of research. However, such rationale behind the choice of CA-terms was often implicit. We need to examine what are the popular CA-terms and their usage contexts in the existing literature.

3 METHODOLOGY

Below, we walk through the data collection process and the procedures for answering the research questions.

3.1 Data collection

We chose the Web of Science Core Collection (WoSCC) based on our research questions and designed methods. Compared to specialized, archival databases such as ACM Digital Library and IEEE Xplore, WoSCC is a multidisciplinary database with a long history, arguably being the largest, most comprehensive, and the oldest subscription-based citation indexing database of peer-reviewed journals and conference proceedings [36, 94]. WoSCC covers over 21,000

peer-reviewed journals and over 300,000 conference proceedings including SIGCHI conferences like ACM Conference on Computer-Supported Cooperative Work and Social Computing and ACM CHI Conference on Human Factors in Computing Systems [20, 94]. Besides, WoSCC supports bibliometric analysis by allowing to export the full records with cited references. Researchers can easily use bibliometric software to conduct co-citation analysis based on the collected data. Since the earliest work can be traced back from WoSCC in 1990, we retrieved CA-related literature dating from 1990 to 2022 from WoSCC.

3.2 RQ1: What are the different terms used to refer to conversational agents in the literature?

3.2.1 Sampling the Most Influential Papers For CA-Terms Identification. On January 10, 2022, we conducted an initial search through WoSCC, with a search query *conversation* agent** in *all fields*, and collected 5,047 papers, dating from 1991 to 2022. The star sign “*” in a search query, as a wildcard character, matches any number of characters in a search term, allowing for more flexible and expansive search results. Specifically, *conversation* agent** will get results containing words such as ‘conversational agent’, ‘conversational agents’, ‘conversation agent’, and ‘conversation agents’.

To identify the most influential papers across CA areas, we conducted a co-citation analysis of the cited references through the CiteSpace software [18]. The CiteSpace software is one of the most comprehensive science mapping software tools for bibliometric analysis [21]. Co-citation analysis, as a type of bibliometric analysis, focuses on publications that are connected when they co-occur in the reference list of a subsequent publication, assuming that publications cited together frequently are similar thematically [92].

The result of co-citation analysis is a network of several co-citation clusters. Unlike other quantitative-oriented approaches to investigate the literature [26], which clustered literature to find patterns based on the assigned numeric data to the selected papers, co-citation clusters are generated by the objective indexes of co-citation relations. For instance, papers A_i and A_j are cited together by a paper while papers B_i and B_j are cited by another paper. Then A_i and A_j , B_i and B_j will be connected respectively through links to present their co-citation relationships. The clusters will then be developed based on the co-citation connections. Papers in the same cluster will most likely contribute to one research area while papers in different clusters will usually contribute to different areas. The co-citation analysis helps to find the most influential publications in a research field [28].

The result identified 1,591 cited references forming 222 clusters. These papers were the most influential papers due to the most frequently cited counts by CA papers. Among the 222 clusters, the top ten largest clusters consisted of 998 papers. To focus on the representativeness of influential papers, we applied the stratified sampling method [78] and chose the top 10% most influential papers references from each of the top ten largest clusters and retrieved 100 references. The 100 papers were cited by a median of 14.5 frequencies (mean=21.51; SD=19.40; range=[4, 108]) dating from 1999 to 2019. Among the 100 references, we excluded 6 references because they were book publications, which might have overlapping

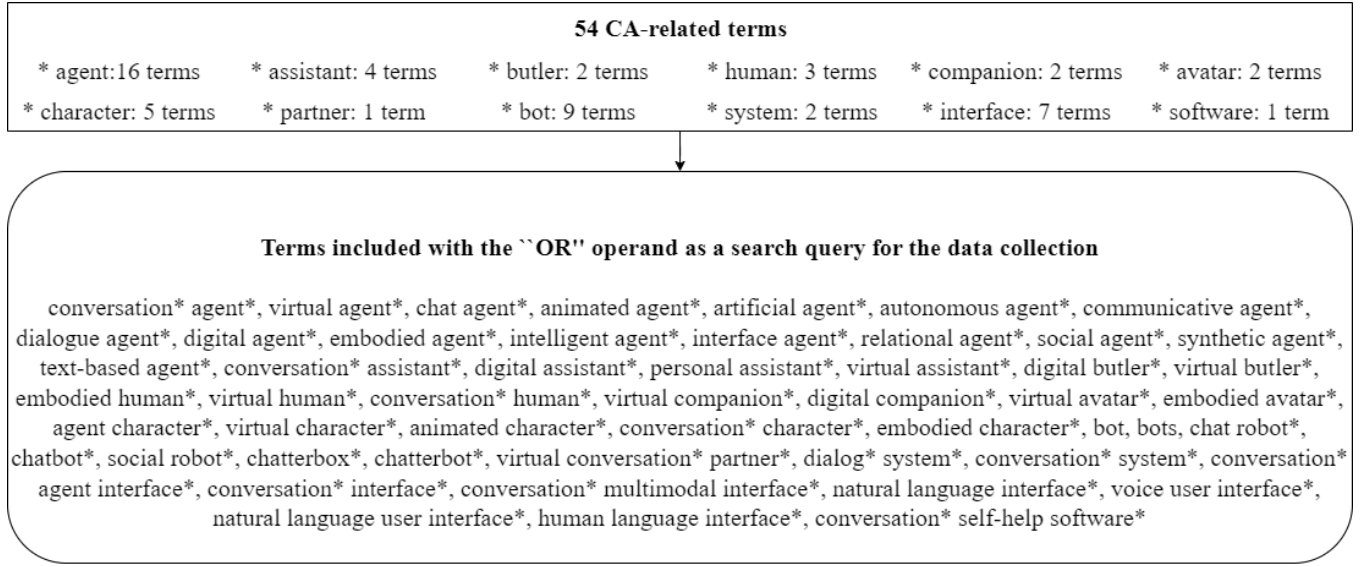


Figure 1: Diagram showing the search query based on the 54 identified CA-terms.

publications with peer-reviewed full publications, leaving a total of 94 references to investigate for extracting CA-terms.

3.2.2 Extracting CA-terms from the literature. We identified CA terms through author-defined terms and additional terms that appeared in the titles, abstracts, keywords, introductions, and related work sections of the 94 references. For the author-defined terms, we collected CA terms if they met two criteria: **(1-1) The authors mentioned the terms as synonyms of CA in their publications.** For example, “*Dialogue systems, also known as interactive conversational agents, virtual agents and sometimes chatbots*” [90]; and **(1-2) The authors of review papers used the terms as search queries** e.g., “*(conversational OR virtual OR digital) AND (agent* OR assistant*) OR chatbot* OR chatterbot* OR chatterbox*” [32].

CA researchers used some terms interchangeably in their writing without either indicating clearly whether these terms were synonyms or providing definitions of these terms. To examine these potential CA terms, we created a codebook of criteria. We consulted a librarian to devise the following inclusion criteria for selecting CA terms. Related terms have to **(2-1) be machine entity, (2-2) interact with users, and (2-3) use natural language as inputs and outputs.** We reviewed the titles, abstracts, keywords, introductions, and related work sections of 94 references to collect CA terms. Examples of excluded terms were: “*human-robot interaction*” since they were not entities of machines (i.e. “*human-robot interaction*” emphasized on “*interaction*” not “*robot*”), “*multi-agent system*” since they referred to interactions between machines or systems, or “*sensitive artificial listener*” or “*rapport agent*” since they focused on non-verbal behaviors. Finally, 23 papers did not have any CA terms, and they either only had terms in technology topics or non-verbal behavior topics or used general terms such as “*robots*”, resulting in 71 papers remaining for examining the usage context of each CA-term.

3.3 RQ2: How do terms on CA and their usage differ depending on their design characteristics and research areas?

3.3.1 Examining Usages in terms of Design Characteristics. To investigate the similarities and differences of the CA-terms, we developed the categories using a qualitative coding process inspired by the thematic analysis [5]. For each reference, we selected relevant statements from the introduction and related work section (quotes) and labeled them as a category of design characteristics of research (code). These steps were performed by one researcher (the first author). After coding all the studies, a second researcher (the third author) reviewed the coding. Both researchers participated in discussion sessions to identify categories that could be merged, renamed, or removed. Finally, we organized the categories by the design characteristics of CAs: (1) text-based, (2) voice-based, (3) non-verbal, (4) capable of being social (e.g., personalities and emotional intelligence with the ability to recognize and display emotions [24, 71]), and (5) remembering the past conversations with each user.

3.3.2 Examining usages in terms of research areas. We identified the research areas of the 71 papers and mapped the areas to the terms used. We extracted the research area labels of these papers from WoSCC and calculated the frequencies of the paper’s research areas if the terms were used in the paper. For example, *conversation* system** had been used in 6 papers. Each paper of these 6 papers according to WoSCC had research area labels with one or multiple research areas. For example, [15] was “Computer Science, Engineering” and [40] was “Psychology”. Thus, for *conversation* system** research area, we calculated the total frequencies of paper for each research area. The two example papers would contribute to added frequencies of “Computer Science”, “Engineering”, and “Psychology”. Finally, the terms with the highest frequencies would be the most used terms.

Table 1: This table shows different CA design characteristics (e.g., text vs. voice) mapped with CA-terms in the retrieved articles. ● means when the term was used in the context of the CA characteristic, ○ means when the term was not used in the context of the CA characteristic, ◐ means when the term was sometimes used in the context of the CA characteristic by those articles. For instance, *chatbots* are text-based but sometimes also voice-based when described as the synonyms of *conversational agents*.

Text	Voice	Non-verbal behavior	Being social	Remembering users' data	Terms
●	○	○	○	○	text-based *, e.g. Messenger, Slack, Telegram
●	◐	○	○	○	chatbot/chat bot/chatterbot/chatterbox/bot
○	●	○	○	○	voice */spoken *, e.g. AT&T's How May I Help You [38]
◐	●	◐	○	○	conversational agent/* assistant/* butler, e.g. Siri, Cortana, Alexa
○	○	●	●	○	sensitive artificial listener/rapport agent (excluded in this study)
●	●	●	◐	○	embodied */animated */multimodal */* human/* character/* avatar/synthetic agent, e.g. SimSensei [27]
●	○	○	●	○	social bot
○	●	○	●	○	social agent
●	●	●	●	○	relational agent
●	●	●	●	●	* companion

Terms were used to refer to any kinds of CAs: dialog/dialogue/conversational system, conversational/human language/natural language interface

4 RESULTS

4.1 The 54 terms that refer to conversational agents (RQ1)

We identified 54 CA-terms and collected 519,513 papers on March 2, 2022, with a combined search query for all fields, shown in Figure 1. This helped us find more than one hundred times of papers searched by previous research query: *conversation* agent**, indicating how important CA research should be aware of such diverse CA-terms if they need to collect as much related literature as possible.

In total, we found 12 different ending words (e.g., XX system vs. XX agent) for labeling CAs. These ending words varied from machine-based entities: *system*, *interface*, *software*, and *bot*; to metaphor-based entities for the meaning of a 'helper': *agent*, *assistant*, and *butler*; or the meaning of a 'person': *human*, *character*, *avatar*, *partner*, and *companion*. On top of these 54 CA-terms, we observed other characterizing adjectives for the agent added as prefixes to those terms: *embodied*, *multimodal*, *text-based*, etc, resulting in 'embodied virtual agent' vs. 'multimodal virtual agent', for instance. Table A1 shows further results.

4.2 CA-terms according to the CA design characteristics (RQ2)

We observed five design characteristics of CAs described in the selected papers, which were shown in Table 1: (1) text-based, (2) voice-based, (3) non-verbal, (4) capable of being social (e.g., personalities and emotional intelligence with the ability to recognize and display emotions [24, 71]), and (5) remembering the past conversations with each user. Some CA researchers applied these terms to refer to these characteristics of CAs while others used these terms interchangeably.

Text-based CAs such as Facebook Messengers, Slack, and Telegram often used terms with *text-based*, leading to varied terms such as *text-based agent* or *text-based conversational interface*. Others used 'chatbot' as a term, and its varied use included *chatbot/chat bot/chatterbot/chatterbox/bot*. Voice-based CAs used *voice user interface* and *spoken dialogue system*. Similar CAs supporting voice and text interaction in Apple's Siri, Microsoft Cortana, and Amazon's Alexa used terms like *conversational agent*. The

terms replacing 'agent' with 'assistant' or 'butler' included *conversational/digital/personal/virtual assistant* and *digital/virtual butler*.

For CAs featured with non-verbal behaviors like gesture, gaze etc., and in examples of SimSensei [27], they applied terms in the 'person-like' metaphor including *human*, *character*, and *avatar* and the terms with restricted words like *embodied*, *animated*, and *multimodal*. Such CAs provided great potential for researchers to explore CAs' social capabilities through verbal and non-verbal channels, which brought a new design characteristic as socializing for disembodied CAs. Research focusing on CAs' social capabilities like personality, emotional intelligence, and communication style [96] to make CAs more socially aware and capable of building good relationships with users, used restricted words 'social' like *social bot*, *social agent*, and *relational agent*.

Unlike CAs with the previous four characteristics, which have no memory or knowledge of users, CAs with the memory of users used 'companion' to refer to this kind of CAs. Such CAs are intended to have long-term interactions and build relationships with users through the knowledge and memory of users' past data and conversation history with the CA. It is also worth noting that these usages were examined based on previous research, and continued to change over time.

The same term (e.g., chatbot) might mean different design characteristics for different researchers. Researchers investigated the opportunities of the disembodied conversational agents which supported both voice and text interactions with the advent of speech recognition technology [3]. Some researchers described *chatbots* and *conversational agents* as the same appellations [22, 95]. Some researchers distinguished *chatbots* and *conversational agents* through other dimensions like *chatbots* were specialized digital assistants and *conversational agents* were general personal assistants[22].

4.3 Disciplinary differences on CA-term use (RQ2)

Based on the calculation of mapped 20 research areas, we showed the most frequently used CA-term for each research area in Table 2. Considering all related research areas, the top five most frequently used terms were 'conversational agent*', 'dialogue/dialog system*',

Table 2: This table shows how different CA research areas use different CA terms.

Research Areas	including */54 terms	The Most used CA-Terms within the areas based on 71 influential papers
Computer Science	46	① conversational agent*, ② virtual agent*, ③ dialogue/dialog system*, ④ chatbot*, ⑤ virtual human*
Engineering	23	① conversational agent*, ② dialogue/dialog system*, ② virtual agent*, ② animated agent*, ② chatterbot*
Psychology	19	① chatbot*, ② conversational agent*, ③ animated agent*, ③ chatterbot*
Health Care Sciences & Services	10	① conversational agent*, ② chatbot*, ② dialogue/dialog system*, ② conversational interface*
Medical Informatics	10	① conversational agent*, ② chatbot*, ② dialogue/dialog system*, ② conversational interface*
Linguistics	7	① conversational agent*, ① chatbot*, ① conversational interface*, ① virtual assistant*, ① conversational humanoid*, ① personal assistant*, ① digital assistant*
Operations Research & Management Science	5	① conversational agent*, ① chatbot*, ① virtual agent*, ① relational agent*, ① virtual avatar*
Psychiatry	5	① conversational agent*, ② chatbot*, ③ conversational interface*, ③ virtual assistant*, ③ text-based agent*
Acoustics	4	① conversational agent*, ① virtual agent*, ① dialogue/dialog system*, ① talkative bot*
Education & Educational Research	4	① conversational agent*, ② dialogue/dialog system*, ② intelligent agent*, ② natural language interface*
Information Science & Library Science	3	① conversational agent*, ② dialogue/dialog system*, ② chatbot*
Public, Environmental & Occupational Health	3	① conversational agent*, ① dialogue/dialog system*, ① relational agent*
Social Sciences - Other Topics	3	① conversational agent*, ① dialogue/dialog system*, ① relational agent*
Automation & Control Systems	2	① dialogue/dialog system*, ① virtual human*
Robotics	2	① dialogue/dialog system*, ① virtual human*
Communication	1	① conversational agent*
Geriatrics & Gerontology	1	① conversational agent*
Mathematics	1	① conversational humanoid*
Medicine	1	① conversational agent*
Telecommunications	1	① virtual agent*

‘chatbot*’, ‘virtual agent*’, and ‘virtual human*’. However, this ranking changed across different areas as Table 2 shows.

We further qualitatively investigated the topics within each research area for the top six research areas in Table 2. Different research areas focused on different CA research topics. For example, in Computer Science and Engineering, the field focused on algorithms and model design and development such as neural conversational model [64, 98], behavioral markup language [97], eye communication in a 3D synthetic agent [81], a CA as museum guide [56], end-to-end dialogue systems [90], animated agents [15, 87], and interactive virtual humans [43]. For Linguistics, it was on language processing like VoiceXML[22], verbal and nonverbal grounding acts design [76]. In Psychology, researchers investigated human factors during interaction with CAs and provided implications for CA design by understanding humans such as human perceptions [3, 37], attitudes [68], behaviors [46, 74], social cues for CAs [32], and enhancing user experience [63]. Health/Medical areas often examined how to apply CAs in health and medical applications (e.g., for medical information [9], mental well-being [48, 70], and in clinical psychology or healthcare context [61, 83]).

5 DISCUSSION

In this study, we provided a list of 54 CA-terms. The terms were used with different meanings for different researchers, and for similar concepts, researchers from different areas used different terms. Below, we discuss the implications of how our findings inform advancing CA research given such inconsistencies and fragmentations of language around CA.

5.1 CA-Terms Reflect the Distinctive Characteristics of CAs

The diverse CA-terms were used in the literature to refer to different design goals, informing the functions and roles of CAs. For example, ‘embodied conversational agents’ were those that interact with users through both verbal and non-verbal behaviors, while ‘text-based agents’ only interact with users through text. McTear classified CAs into five classifications: (1) text-based and spoken dialogue systems, (2) voice user interfaces, (3) chatbots, (4) embodied conversational agents, and (5) social robots and situated agents [71]. These are five distinct traditions in dialogue systems research where communities largely worked independently. Nowadays, there is a trend towards collaborations and building work based on the

achievements of artificial intelligence. Our classifications are similar to McTear's, though, retrieved through a bottom-up approach based on the design characteristics.

The chatbots.org suggested more than nine categories of chatbots, including Web Chat, Messenger Chat, AI Chat, etc. With the design, development, and application of CAs, more CA-terms will arise due to the diversifications of its usage domain and the specialization of its functions. It will continue to be more challenging for CA research with the increase in CA-terms.

5.2 Fragmentation of CA Research

We found a trend toward inconsistent uses of CA-terms. Some researchers interchangeably applied CA-terms which could refer to different kinds of CAs [4, 95]. Meanwhile, researchers in different research areas applied different CA-terms to refer to the same type of CAs. For example, Sarikaya [88] used 'personal digital assistant (PDA)' to describe multipurpose dialogue systems that can answer questions, help with goal-oriented tasks, and engage in casual conversation. Others, in the media, use 'chatbot' to describe this sort of system [71].

Whether we need more unified ways to classify CAs is an open question. However, to conduct CA research, a better understanding of how different fields characterize various aspects of CAs and examining how we can better communicate about the differences will be a critical first step.

5.3 Limitations and Future Work

Through bibliometric analysis and qualitative analysis, we identified the most used CA terms. However, it is important to acknowledge the limitations of our methodology. We only gathered terms from the top 10% most influential papers from the ten largest clusters, which may have excluded relevant terms that are less frequently used or unique to smaller clusters. Additionally, we only collected terms once, and it would be ideal to refine the query through multiple iterations to gain a better understanding. We recommend periodically repeating this type of analysis or conducting it as needed for other evolving topics since the process is repeatable.

For future work, we would like to conduct a thorough analysis of the co-citation clusters to understand how existing literature connected with each other and disconnected, and what are the opportunities for future cross-multidisciplinary CA research.

6 CONCLUSIONS

By examining the CA-terms from influential papers across different disciplines, we contributed to a list of CA-terms for CHI researchers to examine and incorporate into their future literature search and knowledge-gathering processes. We also presented the CA-terms frequently used in the contexts of CA design characteristics and research areas. Our findings inform researchers to learn possible variations of CA-terms. This contribution, in turn, provides potential future directions and critical perspectives for the CHI researchers to explore existing knowledge on CA design. Further, this study helps us discover CA research that one might not previously have found as relevant.

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A RESEARCH METHODS

A.1 The List of CA Terms Identification

As Table A1 shows, we identified 54 CA terms with 12 different ending words (e.g., XX system vs. XX agent) for labeling CAs. These ending words varied from machine-based entities: system, interface, software, and bot; to metaphor-based entities for the meaning of a 'helper': agent, assistant, and butler; or the meaning of a 'person': human, character, avatar, partner, and companion. On top of these 54 CA terms, we observed other characterizing adjectives for the agent added as prefixes to those terms: embodied, multimodal, text-based, etc, resulting in 'embodied virtual agent' vs. 'multimodal virtual agent', for instance.

Table A1: 54 CA terms with varied ending words in labeling CAs

Related terms with References using them	
system	conversation* system*, e.g., (embodied interactive) conversational systems [13, 15, 40, 65, 100, 101] dialogue/dialog system*, e.g., (spoken/conversational/embodied/multimodal) dialogue system* [7, 9, 10, 12, 13, 27, 40, 44, 45, 60, 61, 66, 69, 89, 90, 100, 102]
interface	conversational agent interface*, e.g., (embodied) conversational agent interface* [13, 89] conversational interface*, e.g., (embodied/text-based) conversational interface* [9, 12, 13, 22, 23, 33, 41, 48] conversational multimodal interface* [13] human language interface* [89] natural language interface* [9, 34, 60] natural language user interface* [34] voice user interface* [82]
software	conversational self-help software* [23]
bot	bot* [10] chat bot* [23] chat robot* [23] chatbot*, e.g., smartphone-based chatbot* [3, 22, 32, 34, 37, 46, 61, 65, 69, 70, 73, 82, 95, 101] chatterbot*, e.g., (embodied/online) chatterbot* [23, 32, 46, 56, 69, 90] chatterbox* [32] social bot* [3] social robot* [80, 84] talkative bot* [6]
agent	animated agent*, e.g., animated (pedagogical/interface) agent* [14, 15, 25, 50, 50, 74, 81, 87, 97] artificial agent* [55, 80] autonomous agent* [14, 69, 81] chat agent*, e.g., (interactive/online) chat agent* [37] communicative agent* [3] conversational agent*, e.g., (embodied/text-based/animated/disembodied/human-like (humanoid)/interactive/multimodal/virtual) conversational agent* [3, 6–9, 12–15, 19, 22, 23, 30, 32–34, 39, 41–43, 49, 52, 53, 56–58, 60–66, 70, 72, 73, 79, 81, 82, 84, 89, 90, 95, 97, 98] dialogue agent* [102] digital agent* [32] embodied agent* [2, 69, 82] intelligent agent* [11, 13, 46, 60, 80, 81, 83] interface agent* [25, 50, 69] relational agent* [7, 73] social agent* [23, 25, 80, 84] synthetic agent* [81] text-based agent* [33] virtual agent*, e.g., (intelligent/embodied) virtual agent* [6, 11, 19, 30, 32, 47, 56, 58, 65, 73, 80, 82–84, 89, 90]
assistant	conversational assistant* [9, 32, 56] digital assistant*, e.g., general/specialized digital assistant* [22, 32, 69] personal assistant*, e.g., (digital/general/specialized /intelligent/virtual) personal assistant* [10, 22, 82] virtual assistant*, e.g., (intelligent/multipurpose) virtual assistant [22, 32, 69, 95]
butler	digital butler* [69]

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Table A1 – continued from previous page

Related terms with References using them	
character	virtual butler* [69, 82]
	animated character* [77]
	embodied character* [25]
	conversational character*, e.g., embodied conversational character* [2, 13]
	virtual agent character* [89]
human partner	virtual character* [30, 53, 83, 89]
	virtual conversational partner* [23]
	conversational humanoid*, e.g., (multimodal) conversational humanoid* [15, 76]
	embodied human* [82]
avatar	virtual human*, e.g., (interactive) virtual human* [14, 27, 30, 43, 68, 81–83, 97]
	embodied avatar* [3]
	virtual avatar* [73]
companion	virtual companion* [69]
	digital companion* [69]